Assignment 1: Deadlocks

This assignment is for self-study purposes and is not to be submitted

1. A system contains **m** identical resources shared among three processes: P_0, P_1, and P_2. Their maximum resource demands are 3, 4, and 6, respectively. For what value of m, deadlock will not occur?
   i. 6
   ii. 8
   iii. 10
   iv. 13

2. In a system where resources are allocated in single units, resource allocation and requests can be represented as a directed graph connecting processes and resources Given such a graph, what is involved in deadlock detection.

3. Consider the following snapshot of a system:

	Allocation				**Max**				**Available**			
	R_0	R_1	R_2	R_3	R_0	R_1	R_2	R_3	R_0	R_1	R_2	R_3
T_0	0	0	1	2	0	1	1	2				
T_1	1	0	1	0	1	7	6	0	1	5	2	0
T_2	1	1	5	4	2	1	5	6				
T_3	0	6	2	2	0	6	4	2				
T_4	0	0	1	1	0	6	5	3				

Answer the following questions using the banker's algorithm:

   i. What is the content of the matrix Need?
   ii. Is the system in a safe state?
   iii. If a request from thread T_1 arrives for (0,4,2,0), can the request be granted immediately?
   iv. If a request from thread T_1 arrives for (0,6,2,0), can the request be granted immediately?
   v. If a request from thread T_2 arrives for (1,0,1,2), can the request be granted immediately?